

Noah Khaloo

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Experience

Research Scientist Intern

August 2026–Present

Meta Reality Labs

Redmond, WA

- Develop evaluation pipelines to characterize speech and audio quality from wearable devices operating in noisy, real-world acoustic environments.
- Create benchmark datasets for evaluating enhanced speech and wearable-device audio across representative acoustic conditions.
- Analyze acoustic feature representations to assess how noise suppression, speech enhancement, and target-speaker amplification preserve voice quality.
- Investigate objective and perceptual evaluation methods for enhanced speech, applying DSP and signal-level analysis to wearable-device audio.

Machine Learning Engineer (Contract)

June 2026–August 2026

Loreo.ai

Remote

- Evaluated pre-trained speaker-diarization models across languages, accents, and noise conditions for AI transcription workflows.
- Created naturalistic speech benchmarks modeling realistic turn-taking, interruptions, and conversational flow.
- Built reproducible evaluation pipelines, benchmarked GPU/CPU performance, and developed fine-tuning protocols for ASR and diarization models.

PhD Researcher

September 2023–Present

UCSD Phonetics Laboratory

San Diego, CA

- Design and validate human-centered frameworks for perceptual and acoustic evaluation of text-to-speech and other voice-AI outputs.
- Investigate speech representations and signal-level properties of synthesized voices, translating findings into evidence-based improvements for audio user experiences.
- Develop open-source Python tools for scalable acoustic feature extraction, digital signal processing, and speech analysis.

AI Trainer (Contract)

June 2024–October 2025

OpenAI

Remote

- Designed research-style prompts across statistics, linguistics, and cognitive science to rigorously test large language model capabilities.
- Validated outputs for factual accuracy, completeness, and clarity; developed research-backed rubrics for comprehension, reasoning, and usability.

Selected Publications

Khaloo, N., Holliday, N., & Creel, S. (2026). "Perceptual and acoustic correlates of racial identity in text-to-speech voices." *Interspeech 2026*. Accepted.

Education

University of California, San Diego

September 2023–Present

PhD, Linguistics

San Diego, CA

University of California, Irvine

September 2017–June 2021

Bachelor of Arts, Psychology

Irvine, CA

Technical Skills: Python, PyTorch, R, C++

Core Concepts: Voice-AI, AI evaluation, speech production, acoustic signal processing, human-computer interaction